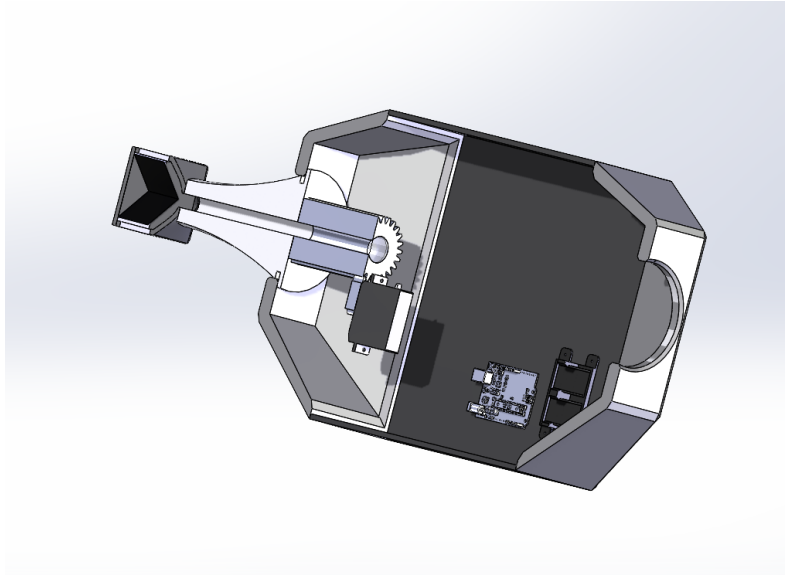


Crazy 8s Balloon Payload

Payload housing and structural design / CU Boulder

For this freshman balloon-payload project, I designed the housing and overall payload structure and contributed structural hand calculations and testing.



Balloon payload housing CAD with internal equipment and an extendable arm

Housing and structural design

An earlier team project involving a balloon-borne payload with an extendable boom arm.

Payload packaging

The team mission called for a balloon payload with an extendable boom arm and imaging equipment. I focused on the housing and overall structure that packaged and supported the payload. The CAD also shows team-developed equipment to explain the packaging context.

Structural work and testing

My contributions included structural hand calculations and testing. The PDR documents planned whip, drop, kick, and cold-environment tests. Those slides record the test plan; they do not by themselves establish that each test passed or that the mission flew successfully.

Team contribution

My ownership was the housing and payload structure. Electronics, sensing, and other mission subsystems were team contributions. This project provides context for the progression toward my later mechanisms, structural-analysis, and flight-hardware work.

Sources: supplied design review or research presentation, with project ownership, test method, and current status clarified by Landon Holligan. Original PowerPoint files accompany the website.